



Australian
National
University

Research School
of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025

Evaporation

EMSC3025/6025

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EMSC3025/6025: Remote Sensing of Water
Resources

Dr. Sia Ghelichkhan

Objectives

By the end of this lecture, you should be able to:

- Define evaporation and its role in the water cycle.
- Differentiate between open water, potential, and actual evaporation.
- Understand transpiration and evapotranspiration.
- Identify key controls: energy, water availability, and vapour pressure deficit.
- Recognise methods to measure and estimate evaporation.

Evaporation

Definition

Transfer of liquid water into gaseous state into the atmosphere

evaporation's importance is mainly affected by two factors:

- 1.
- 2.

Both of these are modulated by climate. Think of cases where evaporation might not be as important:

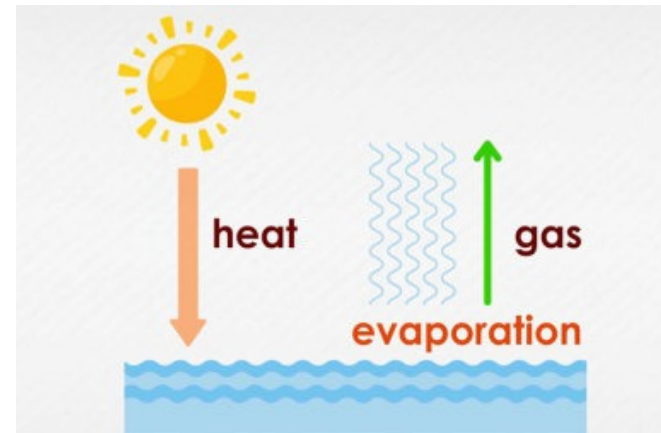


Figure 1: Available water and energy control the main controlling factors

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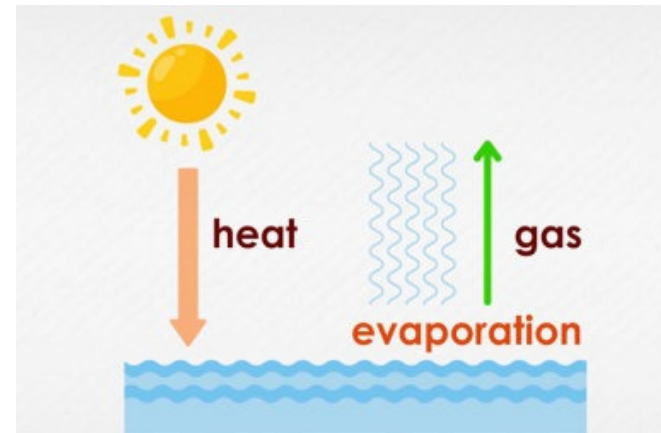


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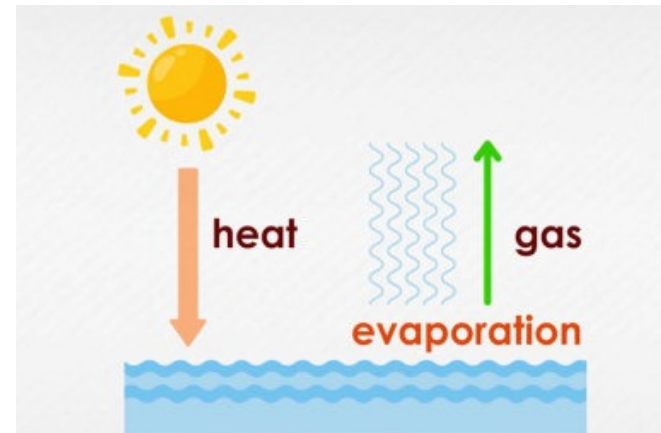


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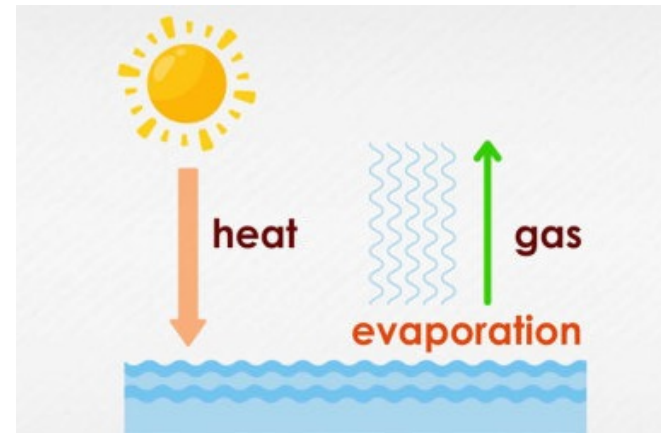


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- Humid climate in winter: much water but **limited energy**

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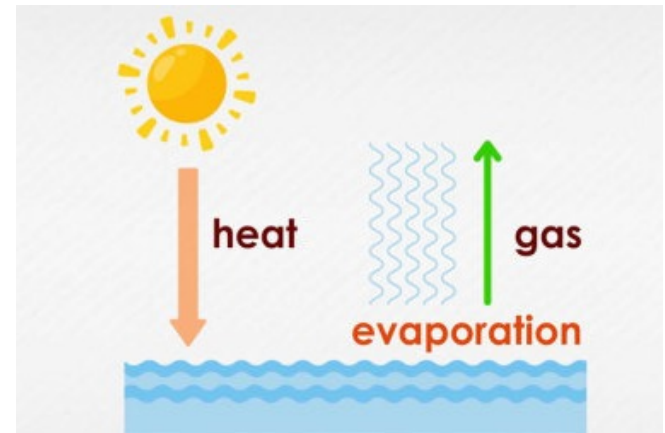


Figure 1: Available water and energy control the main controlling factors

- Humid climate in winter: much water but **limited energy**
- Arid climate in dry season: much energy but **limited water**

Different Definitions of Evaporation

Evaporation

- **Open water Evaporation, E_O** : Evaporation above a body of water such as lakes, streams, oceans. As can be see in global water cycle figure, largest source of evaporation.
 - **Potential Evaporation, E_{pt}** : Amount of evaporation above land if there was unlimited water.
 - **Total Evaporation, B_t** : This is the actual evaporation.
Wet seasons: $E_t \approx E_{pt}$
-
- We are mostly interested in E_O and E_t , but E_{pt} gives us a good indicator of what E_t might be.
 - Fig. 2 → Global Water Cycle, E_O dominates the global component of evaporation

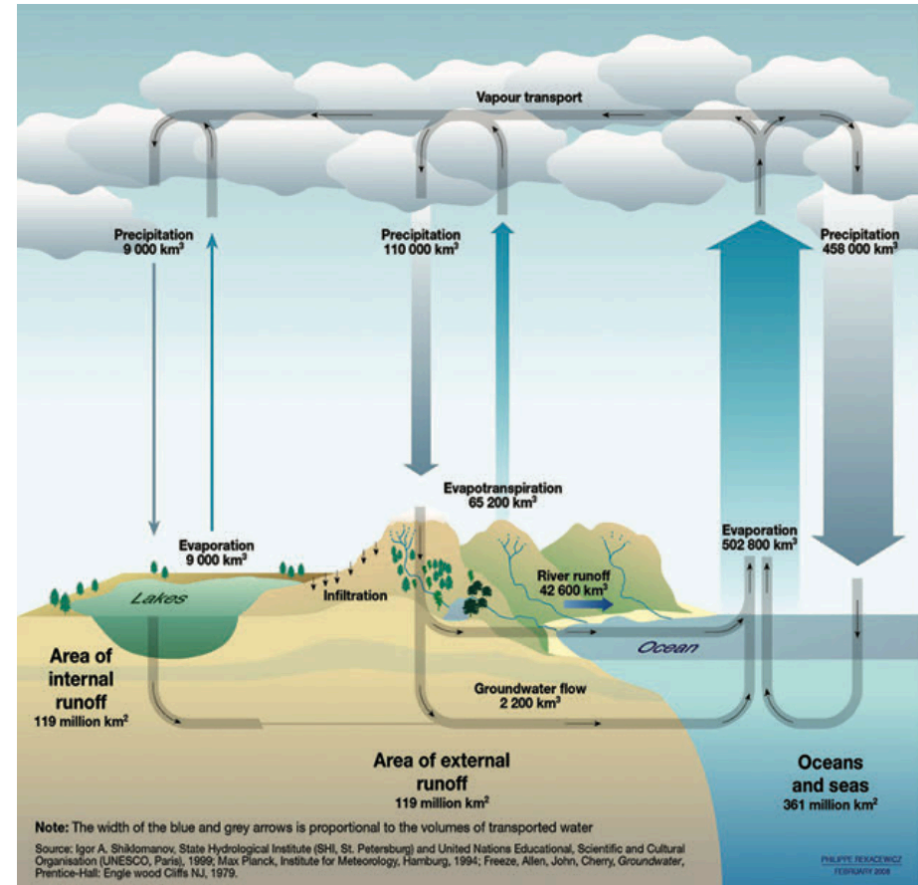


Figure 2: Remember this figure from the water cycle

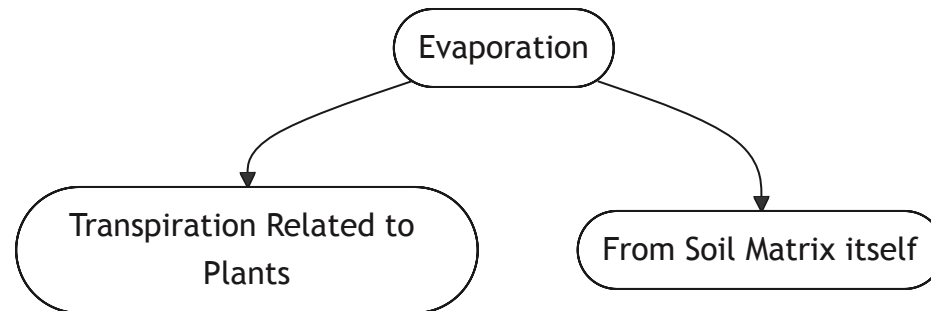


Figure 3: Different Components of Evaporation

Transpiration is the result of photosynthesis and respiration in the plants. It is controlled by:

-
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We are considering evaporation as a negative source. But water lost, is just transferred to the atmosphere. A meteorologist would consider evaporation as a positive source and dewfall as a negative one.

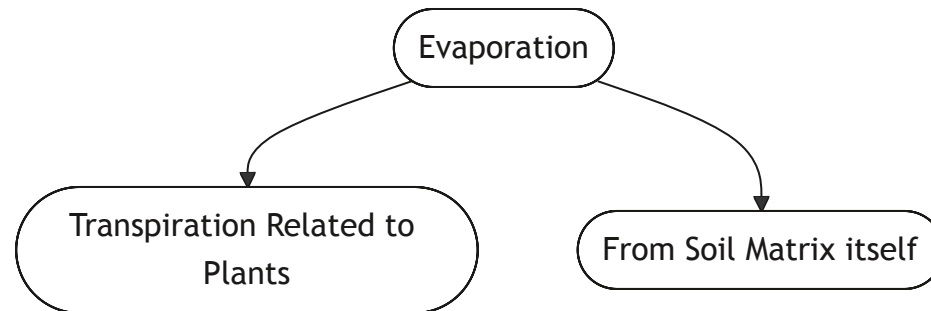


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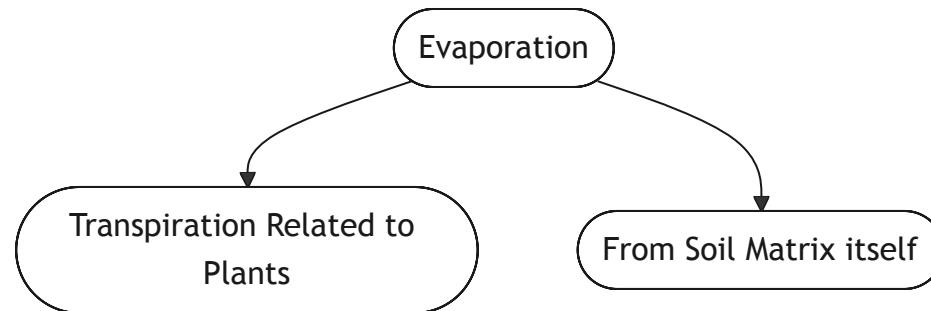


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Transpiration is the result of photosynthesis and respiration in the plants. It is controlled by:

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- ... further by amount of water, the solid, the ability of plants to transfer water to the leaves.

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Transpiration, and bit of fun with TSA

Understanding Transpiration

- **Effect on Local Weather:**

- Increases **heat index**: feels hotter with added humidity.
- High humidity can stress humans by hindering sweat evaporation.
- **Night time temperatures** stay warmer due to high humidity.

- **Seasonal and Regional Impact:**

- Most impactful during hot summer months.
- Affects areas near agricultural hubs like:
 -

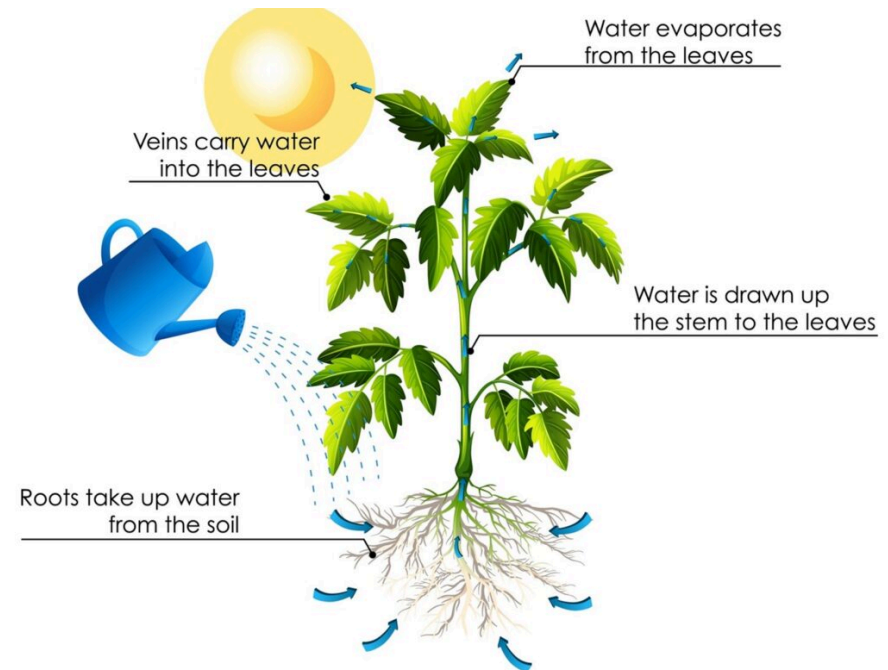


Figure 4: The process of transpiration

Understanding Transpiration

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- Affects areas near agricultural hubs like:
 - cornfields → corn sweat

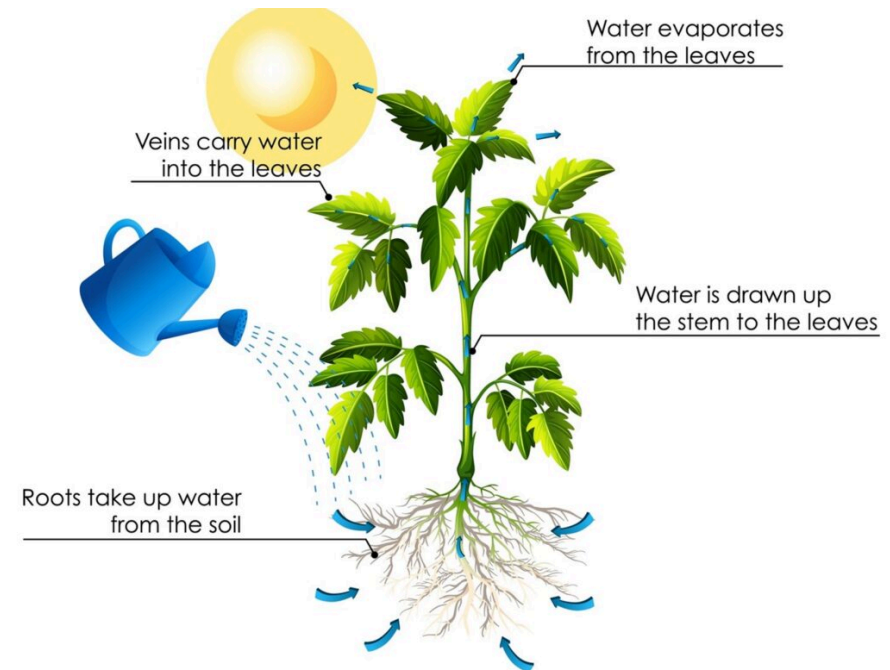


Figure 4: The process of transpiration

Corn Sweat:

A common phenomenon in summer in US midwest

- Millions of acres of cornfields add moisture to the air.
- One acre can release 3,000-4,000 gallons of water per day.
- Iowa's cornfields could add 40.5 billion gallons of water daily.

Evaporation as a process

Dalton (1766-1844): The English physicist was the first to recognise the relationship between wind, air dryness, and evaporation

Formula: $E = C(e_x - e_a)$

Where:

- E — Evaporation rate.
- e_x — Vapour pressure at the water surface.
- e_a — Vapour pressure of the surrounding air.
- C — Empirical constant (depends on wind, temperature, etc.).



Key Idea: Evaporation increases with the difference in vapour pressure between the water surface and the air.

available energy

available water

capacity of air for water

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Available Energy

Main source of energy is from the sun. Sun's radiation is filtered by the clouds, water vapour and absorbed by trees and tall buildings.

$$Q^* = Q_S \pm Q_L \pm Q_G$$

- Q^* : net radiation
- Q_S : Sun's direct radiation (Short wave), heat we feel as warmth
- Q_L : Latent heat due to change in phase. $Q_L < 0$ when Water $\rightarrow$ Gas, $Q_L > 0$ when Gas $\rightarrow$ Water.
- Q_G : Energy absorbed and released from the soil.

Two other sources of energy that *might* become important:

•

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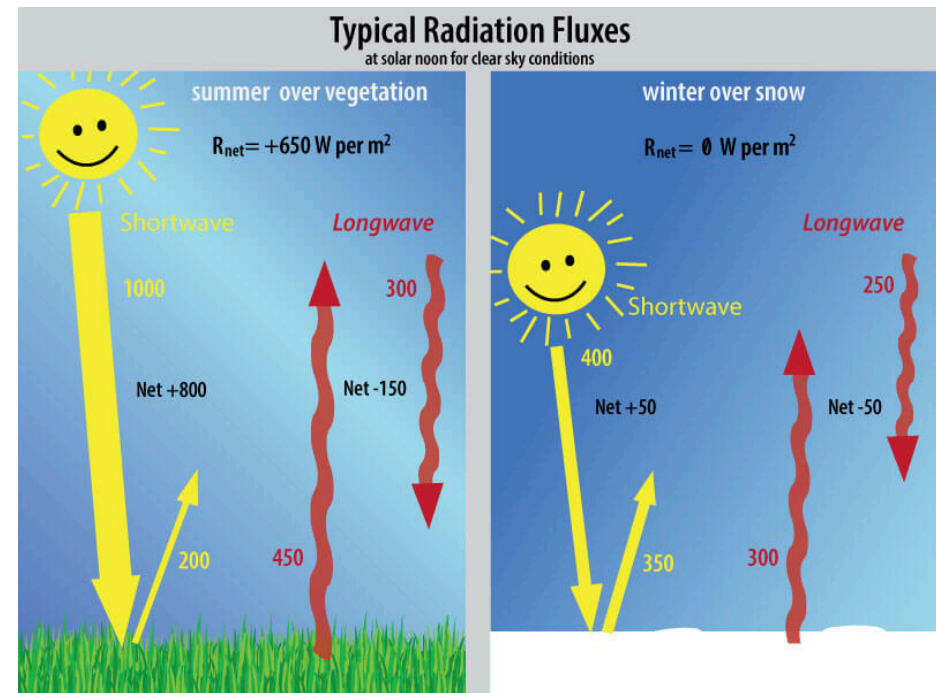


Figure 5

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- In urban settings, energy released from buildings in cold winters. From organic sources of energy.

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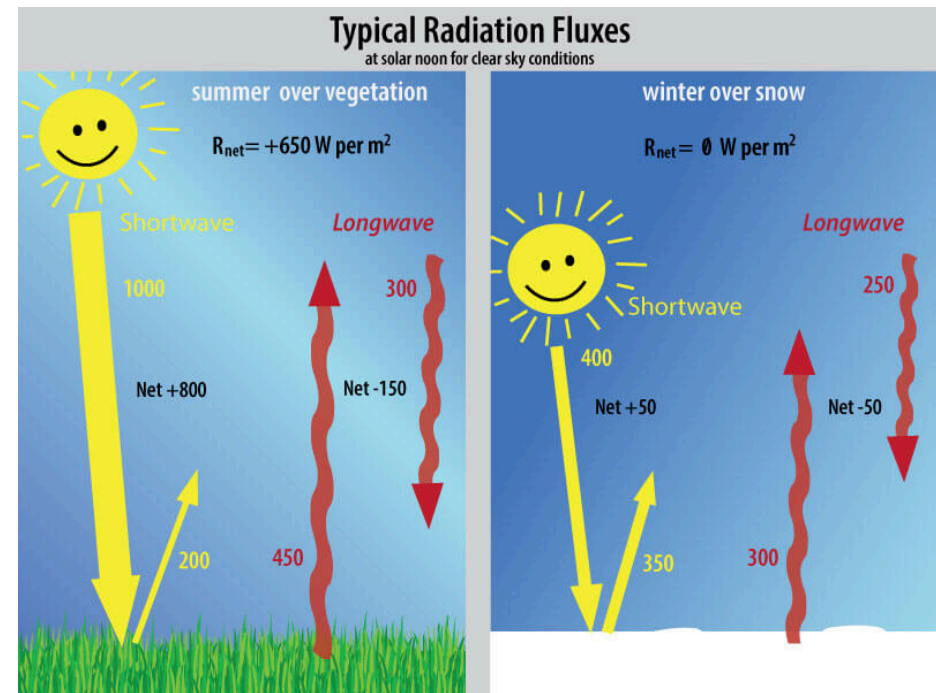


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- Advective heat: As introduced in **introduction**

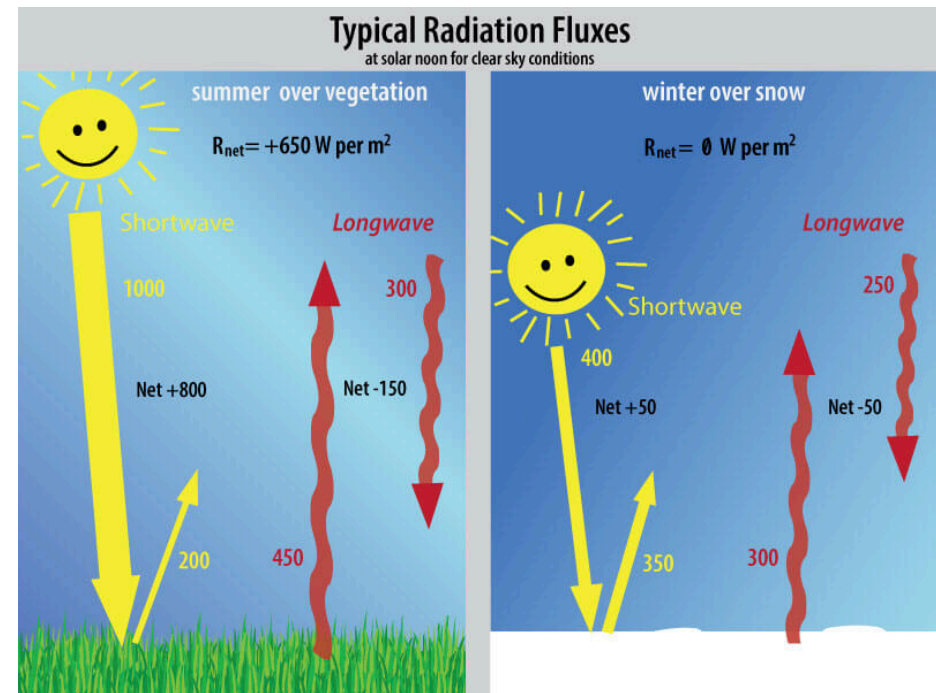
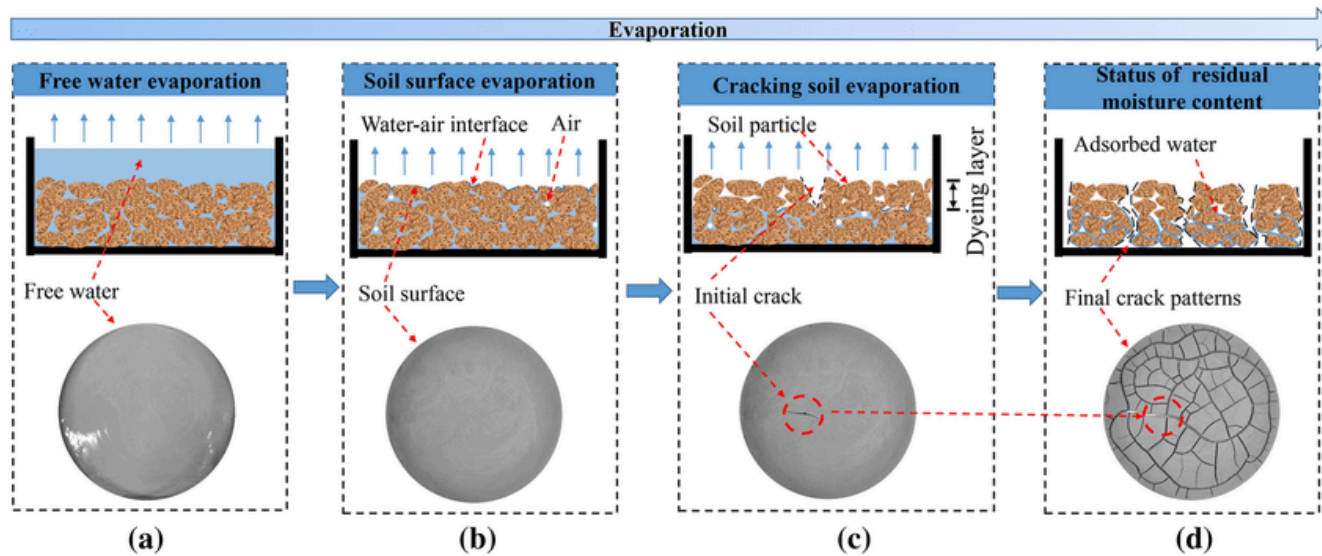


Figure 5

Water supply and evaporation

- Over the **oceans**, water supply is effectively unlimited, making evaporation a major component of the hydrological cycle.
- On **land**, water availability varies:
 - **Open water evaporation** (E_o) occurs from lakes, rivers, ponds.
 - **Soil evaporation** is more complex and limited by water availability.
- In soils:
 - Water evaporates from the **near surface**
 - A **moisture gradient** forms, pulling water upward
 - This upward movement must overcome **gravity** and **soil capillarity**
- Plants also draw water to the surface via **osmosis** in their roots.



Water supply

The receiving atmosphere

- Once the water is evaporated, it diffuses into the atmosphere.
- Evaporation depends on the difference between:
 - e_s : saturated vapour pressure
 - e_a : actual vapour pressure
- The difference ($e_s - e_a$) is the **vapour pressure deficit (vpd)** or **saturation deficit**.
- Higher VPD → higher potential for evaporation
- Relative humidity is the ratio e_a / e_s
- Note that mixing is very important to provide more capacity for evaporation.

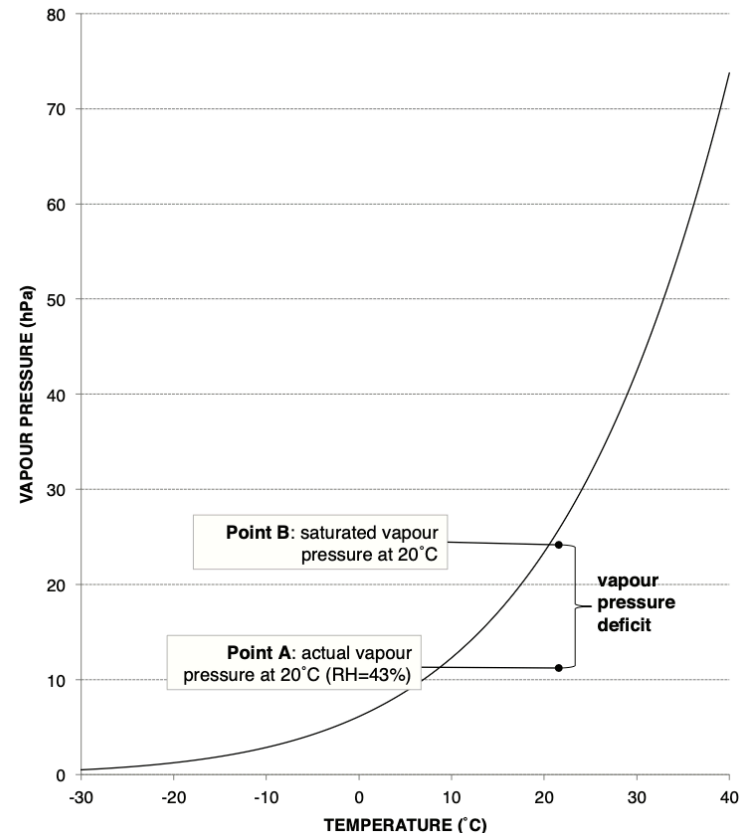


Figure 6: Vapour pressure deficit at 20°C

Evaporation should be looked at in the context of the environment.

Open-water versus soil evaporation

Open water evaporation

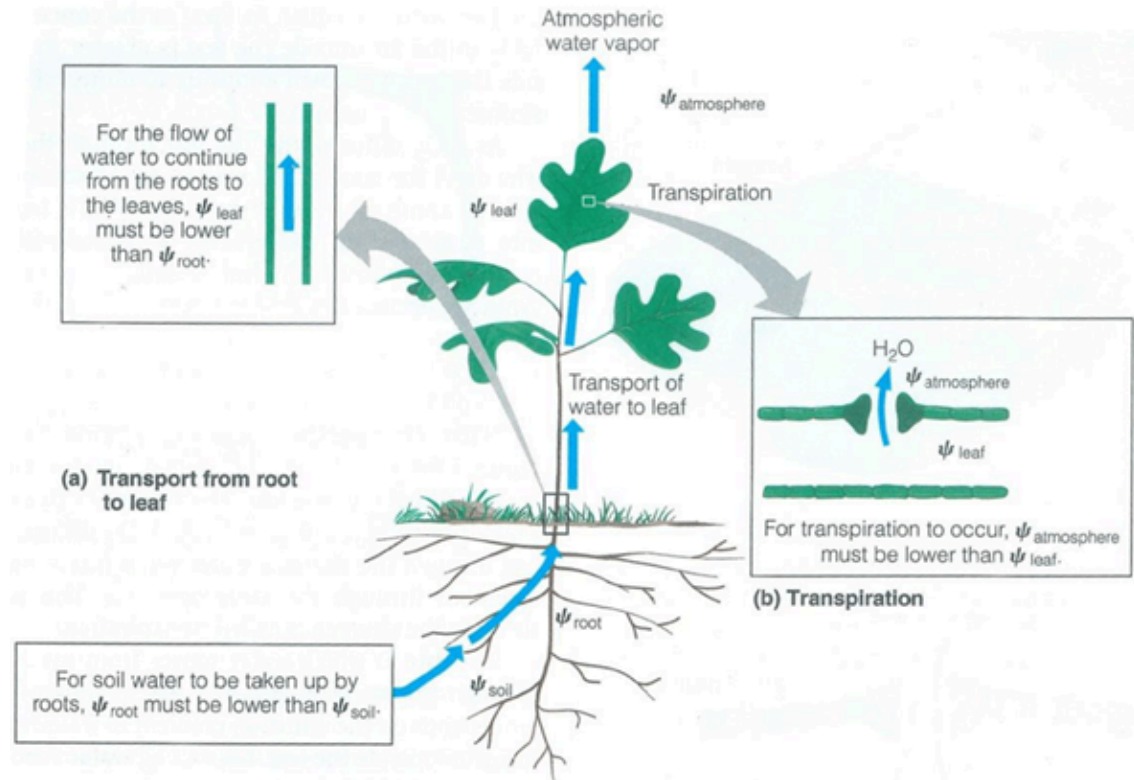
- The simplest case: evaporation from **lakes, rivers, reservoirs**
- Controlled by:
 - Water body properties (temp, salinity, depth)
 - Atmosphere above (vpd and mixing)
- Often **overestimates** evaporation in catchments due to unlimited water supply

Soil evaporation

- Same process as open water, but:
 - **Water supply is limited**
 - Water must move up through the soil
- Influenced by:
 - Soil saturation and capillarity
 - Surface conditions (e.g. bare vs vegetated)

Transpiration and total evaporation

- Transpiration = water loss via **stomata** in leaves
- Plants respond to:
 - Water availability
 - VPD (they close stomata under stress)
- **Evapotranspiration (ET)** combines:
 - Soil evaporation
 - Transpiration
 - Canopy interception loss



Transpiration and total evaporation

Component	Puruki (Central North Island, NZ)(% of annual rainfall)	Balmoral (Central South Island, NZ)(% of annual rainfall)
Annual rainfall	1,405 mm	870 mm
Annual interception loss	370 mm (26%)	220 mm (25%)
Annual transpiration	705 mm (50%)	255 mm (29%)
Annual soil evaporation	95 mm (7%)	210 mm (24%)
Remainder (runoff + percolation)	235 mm (17%)	185 mm (21%)

The relative importance of each component depends on the climate, degree of vegetation cover.

Source: Comparison of two Pine Radiata Forests in New Zealand (Kelliher and Jackson, 2001)

How do we measure evaporation?

Measuring evaporation is hard

- No gauge can directly capture **evaporated vapour**
- Many methods estimate evaporation **indirectly** via:
 - Energy balance
 - Water balance
 - Micro-meteorological measurements
- All methods make assumptions about **fluxes and gradients**

Direct micro-meteorological methods

- Require **high-frequency measurements**, since wind speed, humidity, and temperature are highly variable
- Three techniques:
 - Eddy covariance
 - Aerodynamic profile
 - Bowen ratio

Direct: Eddy covariance

- **Eddy Covariance (Turbulence) Method**
- **Principle:** Directly measures turbulent fluxes by computing the covariance between vertical wind velocity and scalar quantities (e.g., temperature or vapor).
- **Data Needs:** High-frequency 3D wind and scalar measurements.
- **Assumptions:** Stationarity, homogeneous surface.
- **Pros:** Most accurate and direct method; no need for gradient assumptions.
- **Cons:** Expensive; complex instruments and processing; sensitive to sensor alignment, data gaps, and surface heterogeneity.



Figure 7: Eddy covariance system consisting of an ultrasonic anemometer and infrared gas analyser. The anemometer measures the wind speed and direction, and the infrared gas analyser measures the water vapour concentration.

Direct: Aerodynamic profile (Flux-Gradient)

- **Aerodynamic Profiling (Flux-Gradient) Method**
- **Principle:** Relates vertical gradients of wind, temperature, and humidity to turbulent fluxes.
- **Key Inputs:** Profiles of wind speed, temperature, humidity; roughness length; stability corrections.
- **Pros:** Useful in many micrometeorological settings.
- **Cons:** Requires accurate profile data; sensitive to stability corrections and roughness parameters.

Direct: Bowen ratio

- **Bowen Ratio Method**
- **Principle:** Uses measured temperature and humidity gradients between two heights to infer latent and sensible heat fluxes.
- **Key Variable:** Bowen ratio $\beta = H/LE$, where H is sensible and LE is latent heat flux.
- **Data Needs:** Two levels of temperature and humidity, net radiation, ground heat flux.

$$B = \frac{\gamma \cdot \Delta T}{\Delta e}$$

with γ the psychrometric constant.

- Assumes steady state, negligible horizontal advection, similarity of turbulent transfer of heat and moisture.

Indirect: water balance methods

- Evaporation can be inferred from the **water balance**:

$$E = \Delta S - P$$

- Use a pan or lysimeter where:
 - ΔS = change in water volume
 - P = precipitation
- Good for **controlled setups**, less reliable in field conditions

Evaporation pan

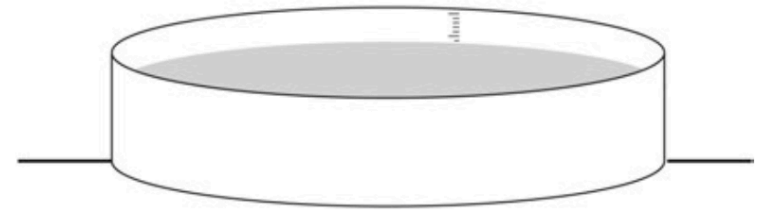


Figure 8

Evaporation pans

- Large containers filled with water, exposed to the atmosphere
- Record water loss over time
- Limitations:
 - Overestimate actual E
 - Suffer from **edge effect** and radiation bias
 - Used mostly for **open water** or **reservoirs**

Evaporation pan

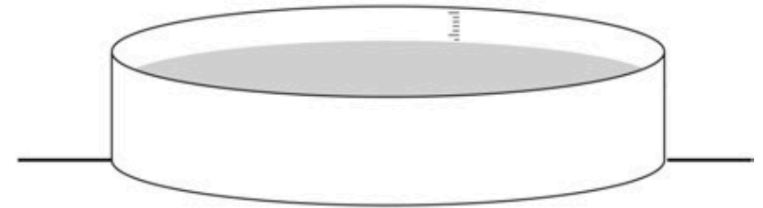


Figure 9: An evaporation pan. This sits above the surface (to lessen rain splash) and has either an instrument to record water depth or a continuous weighing device, to measure changes in volume.

Lysimeters

- Measure E over **vegetated surfaces**
- Mimic soil and vegetation — allow **percolation**

- Use:

$$E = \Delta S - P - Q$$

where Q = drainage

- Can be weighed or monitored using load cells
- Most accurate field method, but:
 - Small scale
 - Expensive and complex

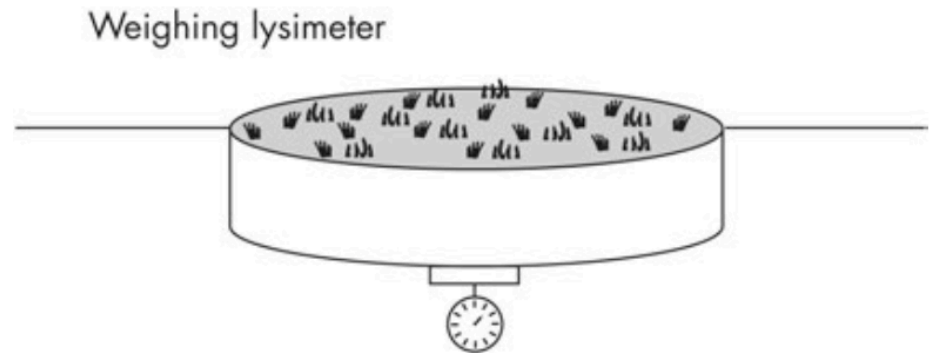
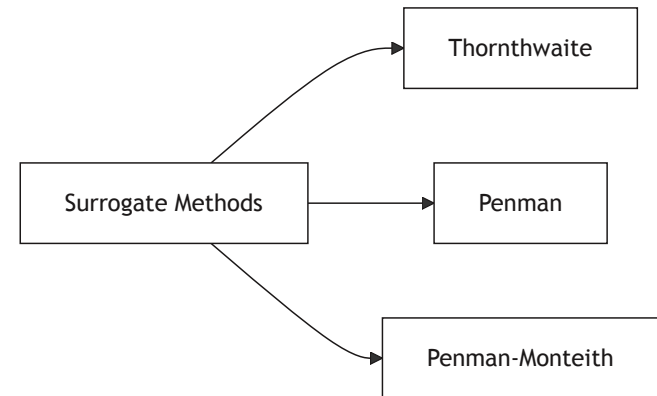


Figure 10: A weighing lysimeter sitting flush with the surface. The cylinder is filled with soil and vegetation similar to the surroundings.

Surrogate methods

- **Surrogate methods** are indirect methods that estimate evaporation based on other variables.
- They focus on climatological variables that are understood to affect evaporation.



Thornthwaite

- A empirical method that estimates potential evaporation based on average temperature temperature.
- Sensible as we know temperature is a proxy for the available energy.
- A monthly heat index (i) for a region

$$i = \left(\frac{T}{5} \right)^{1.514}$$

- Annual heat index (I) is the sum of the monthly heat indices.

$$I = \sum_{i=1}^{12} i$$

- Potential evaporation (E_p) is then given by:

$$E_p = 16b \left(\frac{10T}{I} \right)^a$$

- b accounts for unequal day length between months.
- a is calibrated as a cubic function of the heat index.

$$a = 6.75 \times 10^{-7} I^3 - 7.7 \times 10^{-5} I^2 + 0.018 I + 0.49$$

A **raw estimate** only on temperature.

It can only work on a **monthly scale**. Calibrated for a specific region. For regions with high potential evaporation, **underestimates** evaporation.

Penman for Open Water Evaporation

- Combination of [energy balance](#) and [vapour pressure deficit](#).
- It is a very complex model, but it is very accurate (10 days to daily).

$$E_O = \frac{\Delta Q^* + \gamma E_a}{\Delta + \gamma}$$

where an empirical relationship states that:

$$E_a = 2.6\delta_e \left(1 + \frac{u}{1.862}\right)$$

with Q^* the net radiation, δ_e the vapour pressure deficit, and u the wind speed at 2 m elevation, Δ is the rate of increase of the saturation vapour deficit with temperature, γ is the psychrometric constant (0.063 kPa / °C).

Requires observations of [temperature](#), [wind speed](#), [vapour pressure](#) and [net radiation](#), and provides the rate of evaporation in mm / day. Ignores soil heat flux, and that the vapour pressure deficit is the only limiting factor.

Simplifications to Penman

- Several attempts have been made to simplify the Penman equation for widespread use.
- Priestly and Taylor (1972) derived a simplified Penman formula for large-scale estimation of evaporation, where large-scale advection is not important, and only accounted for through **sensible heat**.

$$PE = \alpha \frac{\Delta(Q^* - Q_G)}{\Delta + \gamma}$$

- Q_G is the soil heat flux term
- α is the Priestly–Taylor parameter, approximates the sensible heat transfer function, estimated by Priestly and Taylor (1972) to be 1.26 for saturated land surfaces, oceans, and lakes.
- This adaptation is important for regional scale evapotranspiration assessment using satellite remote sensing.

Relationship between Actual and Potential Evaporation

- Without stomatal control (e.g., pastures), the relationship between actual evaporation (E_t) and potential evaporation (P_E) is primarily driven by water availability.
- Water availability can be estimated from soil moisture content.
- No units on the x-axis and two different curves (depends on soil and plant types).
- This simple relationship is useful for soil water budgeting models (e.g., Davie et al. 2001) but not for precise modeling.
- The relationship is not linear, and it is not a simple function of soil moisture.

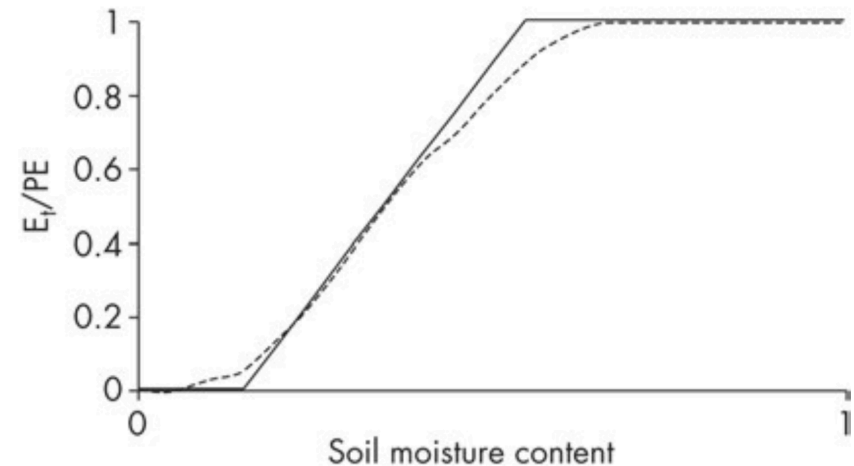


Figure 11: Relationship between actual and potential evaporation

Influence of Vegetation on Evaporation

- In vegetation types with stomatal control (e.g., coniferous forests), evaporation is influenced by both soil moisture and vapour pressure deficit (VPD).
- Figure → shows a time series of soil moisture, transpiration, and VPD for *Pinus radiata* in New Zealand.
- Transpiration was measured using sap-flow meters, soil moisture with a neutron probe, and VPD from a meteorological station.

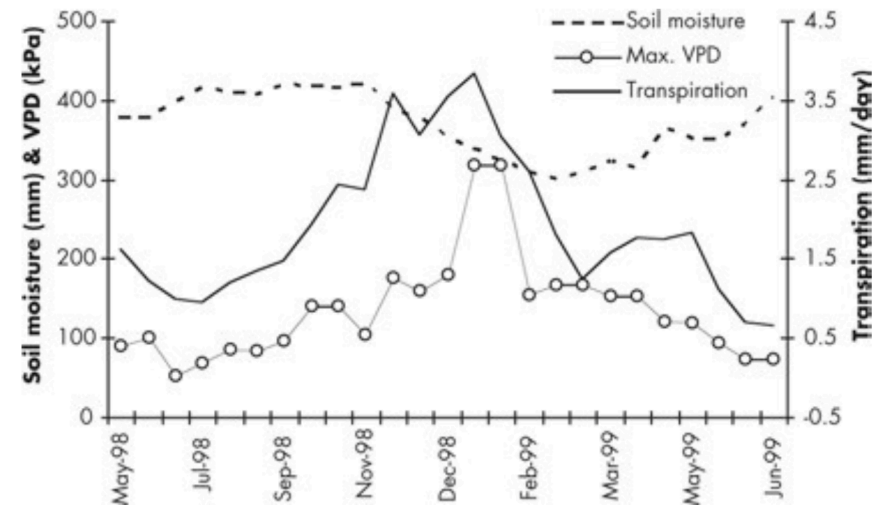


Figure 12: Relationship between actual and potential evaporation

Influence of Vegetation on Evaporation

- At the start of summer (Oct-Nov 1998), high soil moisture leads to a rapid increase in transpiration.
- Transpiration plateaus despite increasing VPD, indicating stomatal control.
- From Jan 1999, transpiration drops, initially matching a drop in VPD, but continues to decrease due to soil moisture limitations.
- Figure 13. → illustrates the complex interaction between evaporation, soil moisture, and atmospheric conditions.

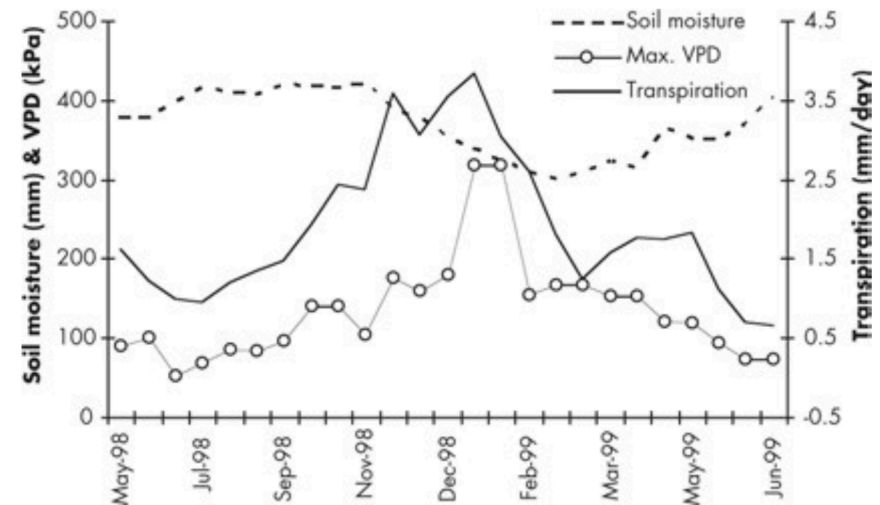


Figure 13: Relationship between actual and potential evaporation

Summary

So here we covered that:

- Evaporation transfers water to the atmosphere, driven by energy and water availability.
- Types include open water (E_o), potential (E_{pt}), and actual (E_t).
- Transpiration from plants adds to total evapotranspiration.
- **VPD** (vapour pressure deficit) is a key driver of evaporation.
- Measured via *direct*(lysimeters) and indirect (pans, water balance) methods.